

CS 505 Module Five Activity Combinatorics Template

Complete this template by replacing the bracketed text with the relevant information.

1. Explain the number of different ways 10 people can be arranged in a line.

Answer:

This question is essentially asking how many permutations there are of 10 people standing in a line. To find the number of permutations, we can use $10!$ (10 factorial) since the number of people decreases after someone is assigned a position in line.

- $10! = 3,628,800$ different ways 10 people can be arranged in a line

2. Explain the number of different ways a committee of five people can be selected from a group of 20 people.

Answer:

Typically, a permutation formula $\frac{n!}{(n-k)!}$ would be used to calculate this if there were specific positions such as president, vice president, etc. Because this is a committee of equal members and specific order does not matter, the formula needs to be modified to the binomial coefficient formula $\frac{n!}{k!(n-k)!}$, where n is the number of people to choose from and k is the number of positions being held.

$$\frac{n!}{k!(n-k)!} = \frac{20!}{5!(20-5)!} = \frac{20!}{5! \times 15!} = \mathbf{15,504 \text{ different ways.}}$$

3. Explain the number of different ways two cards can be selected from a standard deck of 52 cards.

Answer:

Once a card is drawn, that same card cannot be drawn in a standard 52-card deck.

Because of this, the rule of products is used for 2 selections:

$$52 * 51 = 2652.$$

If the order of drawing the cards matters, there are **2,652 different ways** that two cards can be selected from a standard 52-card deck.

If the order does not matter:

$$\frac{2652}{2!} = \frac{2652}{2} = 1,326$$

There are **1,326 different ways** that two cards can be drawn from a standard 52-card deck.

4. Explain how many different 6-letter words can be formed using the letters of the word "MISSISSIPPI."

Answer:

We first need to identify how many choices there are and then calculate the permutations for each choice. I will be using numbers to represent the letter combinations:

1. $4\ 2 = \frac{6!}{4! \times 2!} = \frac{720}{48} = \mathbf{15}$
2. $4\ 1\ 1 = \frac{6!}{4! \times 1! \times 1!} = \frac{720}{24} = \mathbf{30}$
3. $3\ 3 = \frac{6!}{3! \times 3!} = \frac{720}{36} = \mathbf{20}$
4. $3\ 2\ 1 = \frac{6!}{3! \times 2! \times 1!} = \frac{720}{12} = \mathbf{60}$
5. $3\ 1\ 1\ 1 = \frac{6!}{3! \times 1! \times 1! \times 1!} = \frac{720}{6} = \mathbf{120}$
6. $2\ 2\ 2 = \frac{6!}{2! \times 2! \times 2!} = \frac{720}{8} = \mathbf{90}$
7. $2\ 2\ 1\ 1 = \frac{6!}{2! \times 2! \times 1! \times 1!} = \frac{720}{4} = \mathbf{180}$

Combinations per pattern ($2 \geq 4$ letters, $3 \geq 2$ letters, $4 \geq 1$ letters):

$$\binom{2}{1} = 2, \binom{3}{2} = 3, \binom{2}{2} = 1, \binom{3}{3} = 1$$

1. $4 - 2$ choose 1, $2 - 2$ choose 1 = $\binom{2}{1}\binom{2}{1} = 2 \times 2 = 4$
2. $4 - 2$ choose 1, $1 - 3$ choose 2 = $\binom{2}{1}\binom{3}{2} = 2 \times 3 = 6$
3. $3 - 2$ choose 2 = $\binom{2}{2} = 1$
4. $3 - 2$ choose 1, $2 - 2$ choose 1, $1 - 2$ choose 1 = $\binom{2}{1}\binom{2}{1}\binom{2}{1} = 2 \times 2 \times 2 = 8$
5. $3 - 2$ choose 1, $1 - 3$ choose 3 = $\binom{2}{1}\binom{3}{3} = 2 \times 1 = 2$
6. $2 - 3$ choose 3 = $\binom{3}{3} = 1$
7. $2 - 3$ choose 2, $1 - 2$ choose 2 = $\binom{3}{2}\binom{2}{2} = 3 \times 1 = 3$

Total Combinations:

1. $15 \times 4 = 60$
2. $30 \times 6 = 180$
3. $20 \times 1 = 20$
4. $60 \times 8 = 480$
5. $120 \times 2 = 240$
6. $90 \times 1 = 90$
7. $180 \times 3 = 540$

Total: $60 + 180 + 20 + 480 + 240 + 90 + 540 = \mathbf{1610}$

Therefore, there are **1,610** different 6-letter words that can be made from "MISSISSIPPI."

5. Describe real-world applications of combinations and permutations.

Answer:

Real-world applications of combinations and permutations are found in many contexts (possibly just about everything). An example is the many ways to get from one point to another, especially when discussing optimal paths for delivery drivers. Another example might be calculating a password's difficulty based on the available combinations of letters and symbols, and the password length. Next, another example is choosing how many options to display and what to display in a limited space when a company has more products than will fit (e.g., t-shirts, tools, etc.). Lastly, I frequently use combinations and permutations when planning my TTRPG sessions – I need to calculate the various paths my players will take, consider the possible order of players' actions and how I can respond to them, etc.

6. Explain the difference between permutations and combinations.

Answer:

Combinations and permutations both concern the different ways something can be arranged or organized. The primary difference is that permutations rely on a specific order, while combinations do not. In a permutation, changing the order creates a different result. For example, arranging the letters A, B, and C as ABC is different from BAC. In a combination, however, the order does not matter. Selecting two people from a group of five yields the same combination regardless of the order in which they are chosen. Because of this, permutations generally produce more outcomes than combinations.

7. Explain the number of different ways six identical balls can be distributed into six distinct boxes.

Answer:

This problem can be solved using the stars-and-bars strategy and the binomial coefficient formula.

In the stars-and-bars method, the 6 identical balls are represented by stars, and 5 bars are used to divide the balls into 6 distinct boxes. This creates a total of 11 symbols (6 stars and 5 bars).

The number of possible arrangements is calculated using the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{(n-k)! \times k!}$$

Substituting the values into the formula:

$$\binom{11}{5} = \frac{11!}{(11-5)! \times 5!} = \frac{11!}{6! \times 5!} = \frac{39,916,800}{720 \times 120} = \frac{39,916,800}{86,400} = \mathbf{462}$$

Therefore, there are 462 different ways to distribute 6 identical balls into 6 distinct boxes.

8. Explain the number of different ways 15 identical candies can be distributed to 3 children, if each child must receive at least three candies.

Answer:

If every child is expected to receive at least 3 candies, that immediately removes 9 from the options, leaving 6 candies.

Using stars-and-bars, there are 6 stars and 2 bars, resulting in 8 total “variables”.

$$\binom{8}{2} = \frac{8!}{(8-2)! \times 2!} = \frac{8!}{6! \times 2!} = \frac{40320}{720 \times 2} = \frac{40320}{1440} = 28$$

Therefore, there are 28 ways to distribute 15 candies among 3 children, ensuring each gets at least 3.

9. Explain how many non-negative integer solutions there are to the equation $x_1 + x_2 + x_3 = 20$.

Answer:

$$x_i \geq 0$$

$$\text{Using stars-and-bars: } 20 \text{ stars \& 2 bars} = \binom{22}{2} = \frac{22!}{20! \times 2!} = \frac{22 \times 21}{2} = 231$$

Therefore, there are 231 different non-negative integer solutions to the equation:

$$x_1 + x_2 + x_3 = 20$$

10. Explain how many positive integer solutions there are to the equation $x_1 + x_2 + x_3 = 10$.

Answer:

$$x_i > 0$$

Each integer must be at least 1; therefore: $10 - (1 \times 3) = 7$

$$\text{Using stars-and-bars: } 7 \text{ stars, 2 bars} = \binom{9}{2} = \frac{9!}{7! \times 2!} = \frac{9 \times 8}{2} = \frac{72}{2} = 36$$

Therefore, there are 36 different positive integer solutions to the equation: $x_1 + x_2 + x_3 = 10$

11. Describe how the stars and bars method can be adjusted to handle problems with constraints.

Answer:

The stars-and-bars method is extremely useful because it can be adapted to handle different constraints by transforming the variables to satisfy the standard non-negative requirement: $(x_i \geq 0)$.

1. Lower Bound Constraints:

When a problem requires that each group receive a minimum number of items, such as distributing candies to children, we need to adjust the total number of items by subtracting the required minimums (for example, we removed 9 from the 15 candies since each of the 3 kids needed 3). Essentially, the stars get calculated by doing: $n - (k \times x)$.

2. String Positive Constraints

When a problem requires positive-integer solutions, it means no group can be empty. This allows the “gap technique,” resulting in $\binom{n-1}{k-1}$. This is used in question 10, which involves 10 objects and 3 groups, with all groups required to be positive. This gave $\binom{10-1}{3-1} = \binom{9}{2} = \frac{9 \times 8}{2} = 36$

12. Describe real-world applications of stars and bars.

Answer:

There are several different ways that the stars-and-bars method can be applied to real-world scenarios. Below are a few.

1. Finances

Stars-and-bars can be used to find how many budgeting variations someone or a company could have based on income and its division. For instance, if I have \$2000 and need to split it into 4 envelopes or accounts to represent bills, groceries, savings, and fun money, I could calculate the possible ways to split it – including setting lower bounds on how much is required in each account.

2. Game Design & RPGs

In games such as Dungeons & Dragons, this strategy can be used to decide, given stat points, how many combinations those points can be allocated to stats. This helps game developers and people who run these games balance them more effectively.

3. In Computer Science

This strategy can be used to count the number of ways to allocate resources or to distribute tasks among processors.